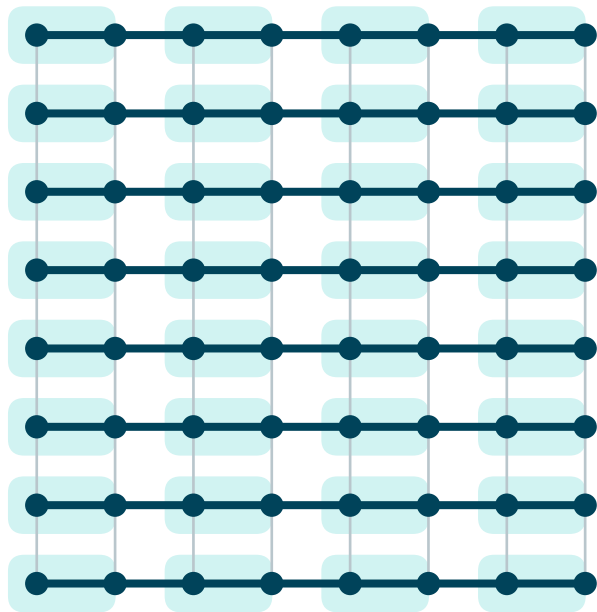
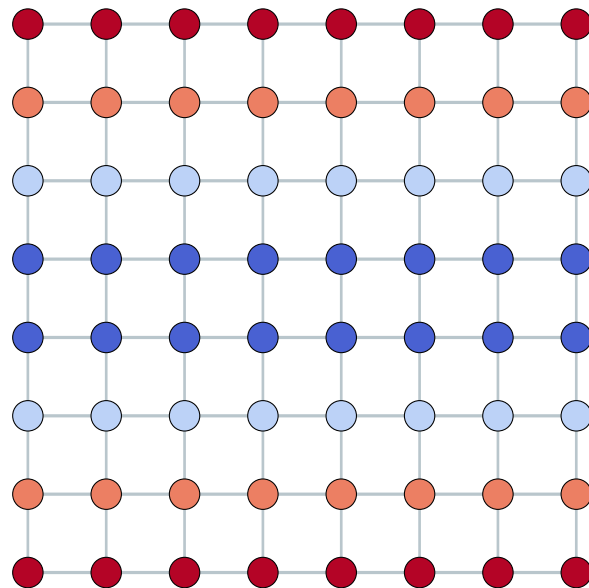


(A) Fine grid + caliber-1 aggregates
strong x (thick), weak $y = \varepsilon$ (thin)



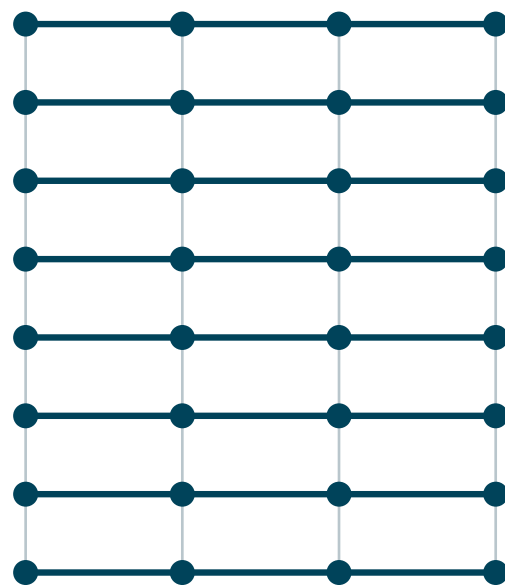
affinity is large across strong x -edges
 $\Rightarrow$ 2:1 pairs along x

(B) Slow mode after pointwise GS
smooth in x , oscillatory in y



GS smooths strong x ; leaves x -smooth error,
 any y -frequency ($\sqrt{n}$ such modes)

(C) Coarse grid = semicoarsened
 $P/2 \times P$: half the x -points, all y -layers



every y -layer kept at full resolution
 $\Rightarrow$ all $\sqrt{n}$ slow modes representable